

In the Name of God



Course Project: Introduction to Machine Learning Phase 2

Bayesian Neural Networks

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Sharif University of Technology

Spring 1404

Deadline:
Tir 10, 1404 at 23:55

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1 Important Notes

Please carefully read the following points:

1. The two phases of this project will contribute a total of **4 points** to your final course grade.
2. After the final submission, there will be an in-person project defense session where you must present your code and outputs to the teaching assistants and answer their questions. **All group members must be fully familiar with every part of the project.** A single score will be assigned to the entire group.
3. All sections of both phases are mandatory. There will be no optional or bonus parts.
4. You are not permitted to modify any predefined sections and code blocks in the notebook.
5. All simulations must be implemented using Python. You are allowed to use any libraries you have used in the assignments, such as **numpy**, **scipy**, and **pytorch**. However, **you must implement the algorithms yourself**, and you may not use libraries that already provide the algorithm implementations.
6. The project deliverables consist of both your code and a written report. The report should include answers to the questions, plots, figures, and conclusions. In the end, one .zip file containing all code files, and a .pdf report must be uploaded to the CW platform. Only one group member needs to submit the project.
7. If you use any external sources (books, articles, websites, etc.) to answer the questions, make sure to cite them properly.
8. **Any form of plagiarism will result in a zero for both parties involved.**
9. For any questions, please contact the project instructors via: @H_Theory, @amir_naghdi_2003, @mh_momeni, @YahyaTehrani, @GhLmu on Telegram.

Good luck!

Introduction to Machine Learning (25737-2)

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2 Introduction

Having established the theoretical foundations in **Phase 1**, this phase shifts focus to the practical application of *Bayesian* methodologies, which rigorously quantify inherent uncertainties in predictive modeling. Phase 2 constitutes a comprehensive, hands-on exploration within a single Jupyter notebook, segmented into four distinct modules, each demonstrating a key facet of Bayesian inference:

1. Bayesian Linear Regression
2. Bayesian Deep Learning via Dropout (Bonus)
3. Basic Markov-Chain Monte Carlo (MCMC)
4. Hamiltonian Monte Carlo (HMC)

This module will illustrate how transitioning from a singular deterministic point estimate to an *entire probability distribution* for parameters profoundly impacts analytical outcomes and decision-making processes.

3 Motivation

The imperative for adopting Bayesian methods stems from the pervasive influence of randomness and inherent complexities in real-world data and systems. Bayesian inference provides a principled framework to *quantify and manage these uncertainties rigorously*, rather than merely acknowledging their existence.

A critical distinction within uncertainty quantification is between *epistemic uncertainty* and *ambiguity*. Epistemic uncertainty refers to the uncertainty arising from a lack of knowledge or insufficient data, which is theoretically reducible with more observations or improved models. Bayesian methods are particularly adept at capturing this type of uncertainty by providing a posterior distribution over model parameters, reflecting the plausibility of different parameter values given the observed data and prior beliefs. Conversely, ambiguity often relates to inherent vagueness or ill-defined aspects of a problem, or structural uncertainty where multiple model formulations might fit the data equally well, making it difficult to distinguish between competing hypotheses even with ample data. While Bayesian methods can sometimes illuminate aspects of model ambiguity through model comparison or multimodel inference, their primary strength lies in addressing reducible epistemic uncertainty.

Illustrative Applications:

- **Medical Electronics.** A Bayesian Neural Network (BNN) tasked with segmenting brain tumors from MRI scans offers more than just a contour; it provides probabilistic confidence maps. For instance, it might indicate "90% certainty in this core region, but only 60% certainty in the peripheral area," thereby empowering radiologists with crucial quantitative uncertainty metrics. Analogously, in wearable ECG monitors, BNNs can flag arrhythmias with associated confidence levels, highlighting beats where the model's predictive certainty is reduced.
- **Critical Digital and RF Systems.** Anomaly detection systems in high-availability network routers or satellite communication links can leverage Bayesian change-point models to generate alerts with *calibrated false-alarm rates*. This ability to quantify the probability of a true anomaly versus a false positive is invaluable for operations, preventing unwarranted interventions and ensuring timely responses to genuine faults.
- **Resource-Constrained Design.** By judiciously employing a set of nonlinear basis functions, a *linear* Bayesian model can be transformed into a computationally efficient predictor. This lightweight approach can often surpass the performance of more complex neural networks while critically providing credible intervals for forecasted load currents, which is vital for robust system design and resource allocation.

Proficiency in these techniques transcends a mere academic exercise; it endows practitioners with a vital capability: the ability to articulate not only "*Here is my prediction*" but also "*Here is the quantifiable confidence associated with this prediction.*" Whether optimizing power converter performance, identifying subtle ECG anomalies, or foreseeing nascent failures in telecommunications infrastructure, this additional dimension of information is pivotal for informed and optimal decision-making.

4 Overview of Notebook Sections

This notebook systematically covers four fundamental Bayesian techniques for data analysis and uncertainty quantification. Each section is designed to provide a deep dive into its respective methodology:

Bayesian Linear Regression This module explores a probabilistic reformulation of linear regression. In contrast to classical linear regression, which yields a single point estimate for parameters based on minimization criteria (e.g., least squares), Bayesian Linear Regression employs Bayes' theorem to coherently integrate prior beliefs (e.g., regarding slope or intercept distributions) with observed data. This integration results in a full posterior probability distribution over possible models, enabling the derivation of credible intervals—a range of plausible predictions rather than a singular forecast. This approach is particularly advantageous in scenarios characterized by scarce or noisy data. The notebook will guide the user through the formal setup of priors, the computation of the posterior distribution, and the visualization of how predictive uncertainty diminishes with the accumulation of additional data.

Bayesian Deep Learning via Dropout This section addresses the integration of uncertainty quantification within deep learning architectures. It specifically examines how dropout, a regularization technique conventionally employed to mitigate overfitting in neural networks, can be reinterpreted as a Bayesian approximation method. Technically, dropout performs an implicit form of variational inference by stochastically deactivating neurons during training, effectively sampling from an ensemble of network configurations. This process yields not merely a point prediction, but also an estimate of predictive confidence—akin to a neural network expressing, "I am 80% confident in this output." A key distinction of Bayesian Neural Networks (BNNs), particularly those employing dropout for uncertainty estimation, is their capacity for correlated predictions. Unlike traditional neural networks that produce independent predictions for distinct inputs given a fixed set of weights, BNNs provide a distribution over weights. When sampled,

this distribution introduces dependencies across predictions for different inputs, as the uncertainty in shared underlying weights propagates consistently. The notebook will elucidate the implementation of this technique, demonstrating how to extract and interpret uncertainty estimates from a trained model, emphasizing its significance in high-stakes applications such as medical diagnostics.

Basic Markov Chain Monte Carlo (MCMC) This module introduces Markov Chain Monte Carlo (MCMC) as an indispensable tool for sampling from complex or intractable probability distributions, particularly when analytical solutions are infeasible. MCMC operates by constructing a Markov chain that traverses the parameter space, eventually converging to a stationary distribution that mirrors the desired posterior distribution. This can be conceptualized as a guided random walk designed to thoroughly explore the landscape of possible model parameters. This section provides a foundational understanding of the Metropolis-Hastings algorithm, a canonical MCMC method, and illustrates its utility in fundamental Bayesian inference problems. The notebook will facilitate the execution of MCMC on a simplified problem, allowing for the visualization of sampled paths and their convergence to the underlying posterior distribution.

Hamiltonian Monte Carlo (HMC) This section presents Hamiltonian Monte Carlo (HMC), an advanced MCMC algorithm that significantly enhances sampling efficiency. HMC draws inspiration from Hamiltonian dynamics in physics, treating model parameters as particles possessing momentum. Rather than relying on purely random steps, HMC leverages the gradient of the posterior distribution to guide its proposals, rendering it substantially more efficient in exploring high-dimensional parameter spaces (e.g., models with dozens or hundreds of parameters). This efficiency is paramount for complex machine learning models. The notebook will introduce the leapfrog integrator—a core numerical method within HMC—and provide opportunities to experiment with tuning its hyperparameters, such as step size, to observe its superior performance compared to basic MCMC.

5 Conclusion

This notebook serves as a crucial bridge from theoretical understanding to practical implementation, offering direct engagement with Bayesian methodologies across four structured sections. By meticulously working through this material, users will not only refine their technical skills but also acquire a profound comprehension of uncertainty management in data-driven contexts. This endeavor represents a significant opportunity to translate abstract concepts into tangible, real-world solutions.

Good luck with your project! May your models be well-calibrated and your uncertainties quantified.

6 References

1. Bayes, T. (1763). An Essay Towards Solving a Problem in the Doctrine of Chances. *Philosophical Transactions of the Royal Society of London*, 53, 370-418.
2. Gal, Y., & Ghahramani, Z. (2016). Dropout as a Bayesian Approximation: Representing Model Uncertainty in Deep Learning. *Proceedings of the 33rd International Conference on Machine Learning (ICML)*, 1050-1059.
3. Metropolis, N., Rosenbluth, A. W., Rosenbluth, M. N., Teller, A. H., & Teller, E. (1953). Equation of State Calculations by Fast Computing Machines. *The Journal of Chemical Physics*, 21(6), 1087-1092.
4. Neal, R. M. (2011). MCMC Using Hamiltonian Dynamics. In S. Brooks, A. Gelman, G. Jones, & X. Meng (Eds.), *Handbook of Markov Chain Monte Carlo* (pp. 113-162). Chapman and Hall/CRC.